Mini Project

Exploratory Data Analysis of House Prices

A practical walkthrough of data inspection, cleaning, summaries, visualizations, findings, and limitations

DATASET	4,600 rows | 18 original columns
PERIOD	02 May 2014 - 10 Jul 2014
GEOGRAPHY	44 cities in Washington
TOOLS	Python | Pandas | NumPy | Matplotlib | Seaborn

Project objective

Describe the supplied home-sale records and explore how sale prices vary with living area, location, and selected home characteristics. This is an introductory EDA project, not a predictive model or a current valuation guide.

Executive summary

The source contains 4,600 rows and no missing cells or exact duplicate rows. It includes 49 non-positive prices (1.1%), which are excluded from price-based summaries. For the remaining sales, the median price is $465,000; the mean is $557,906. Price has a positive correlation (0.45) with living area, while observed city medians vary widely. Short time coverage and limited source documentation constrain interpretation.

1. Data and workflow

The analysis follows a beginner-friendly sequence: load the CSV, inspect its structure, check quality, prepare analysis fields, summarize distributions and groups, visualize patterns, then state findings and limitations.

Data quality and preparation

The file has 4,600 sales, 18 source columns, 0 missing cells, and 0 exact duplicate rows. Sale date is parsed as a date, text values are trimmed, and helper columns capture sale month, home age at sale, basement presence, and renovation presence. The raw input is retained unchanged in the data folder. The cleaned output contains 4,551 positive-price records; 49 non-positive-price records are omitted because they cannot represent an ordinary observed sale price.

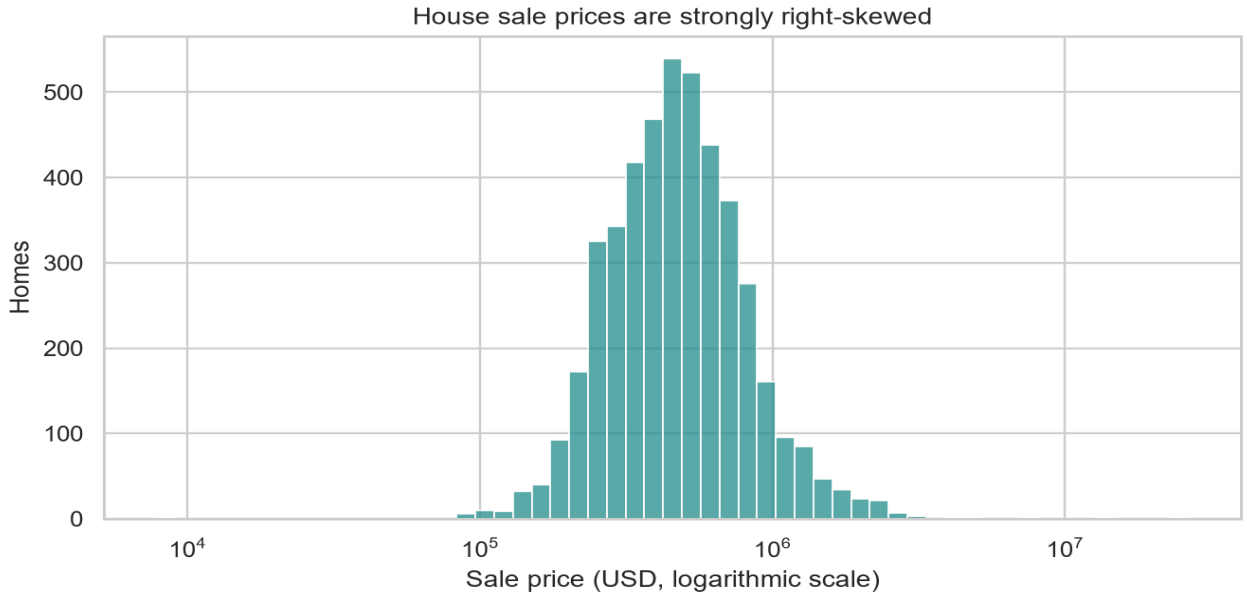
Key descriptive statistics

Measure	Result
Positive-price sales	4,551
Median sale price	$465,000
Mean sale price	$557,906
25th - 75th percentile	$326,264 - $657,500
Highest observed price	$26,590,000
Price / living-area correlation	0.45

Interpretation: the median is a more robust description of a typical sale than the mean because the price distribution has a long upper tail. The maximum is unusually high and should be checked against source documentation before modeling.

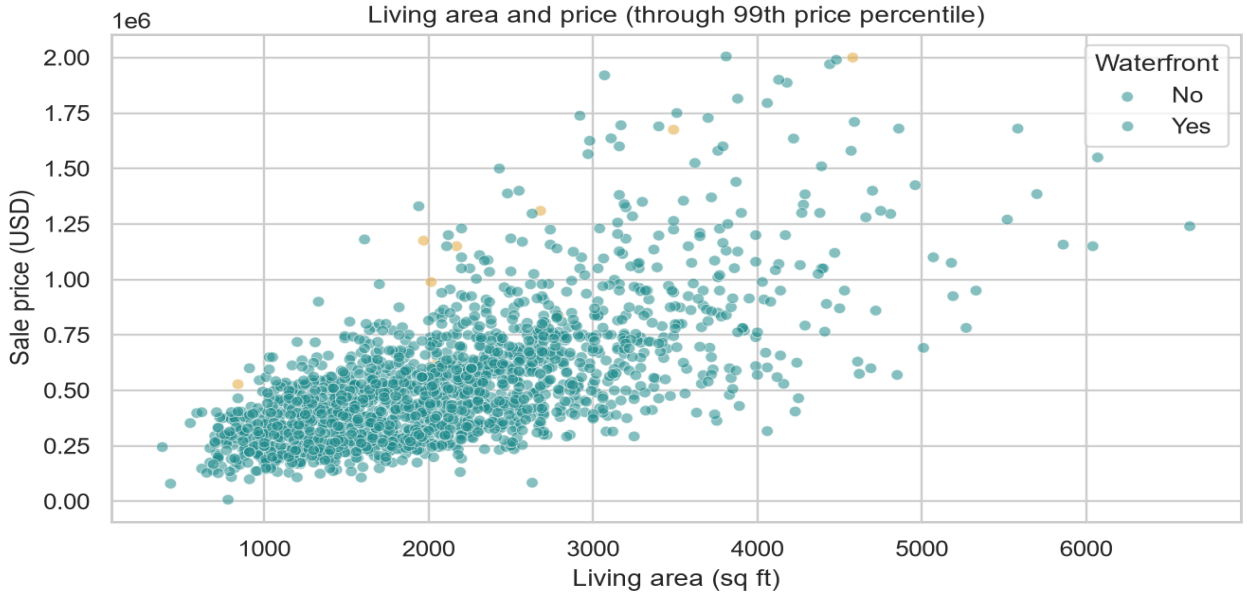
2. Visual exploration

Price distribution



The distribution is strongly right-skewed. A logarithmic horizontal axis makes both the main cluster and high-priced sales visible; no positive-price records were removed from this histogram.

Living area and price



The scatter shows a positive association (Pearson r = 0.45). To make the main pattern readable, the plotted points stop at the 99th price percentile; the full data remain in the cleaned CSV and the extreme values are not deleted. This is association, not a causal effect.

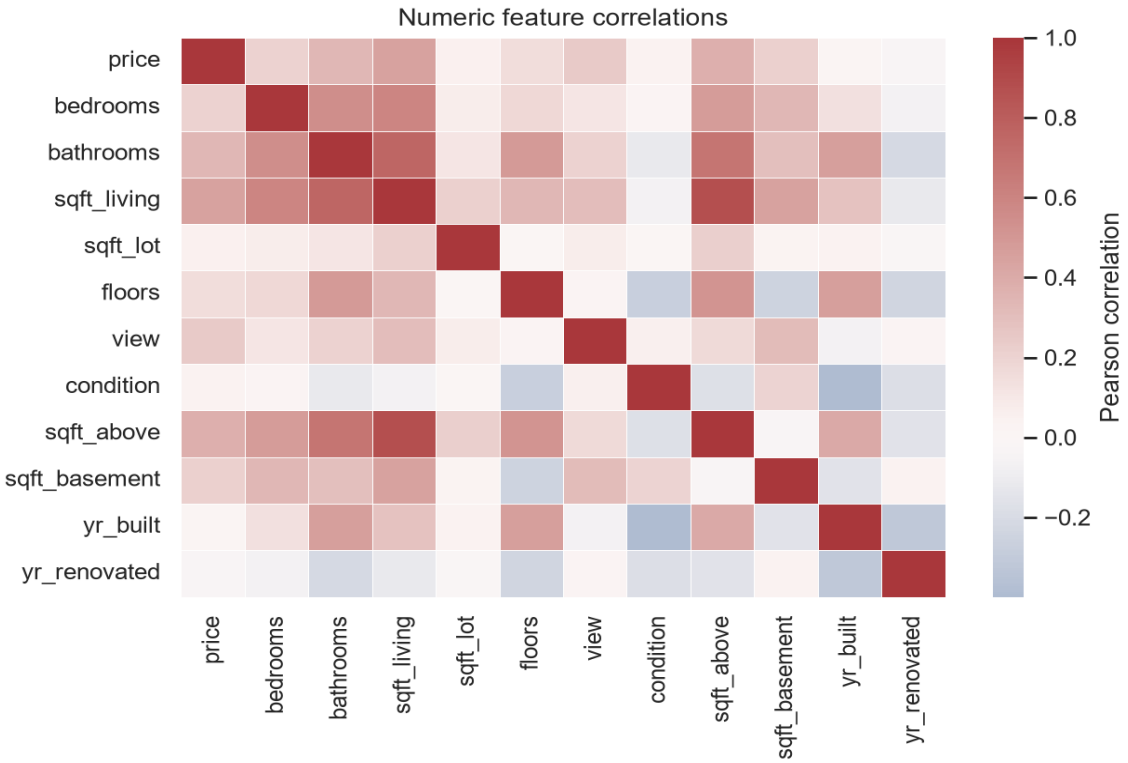
3. Location and feature comparisons

City median sale prices



Only cities with at least 20 positive-price sales are shown. The highest median among these is Mercer Island ($950,467; n=82); the lowest is SeaTac ($238,500; n=28). City medians combine differences in home size, amenities, and neighborhood mix, so they do not isolate a location effect.

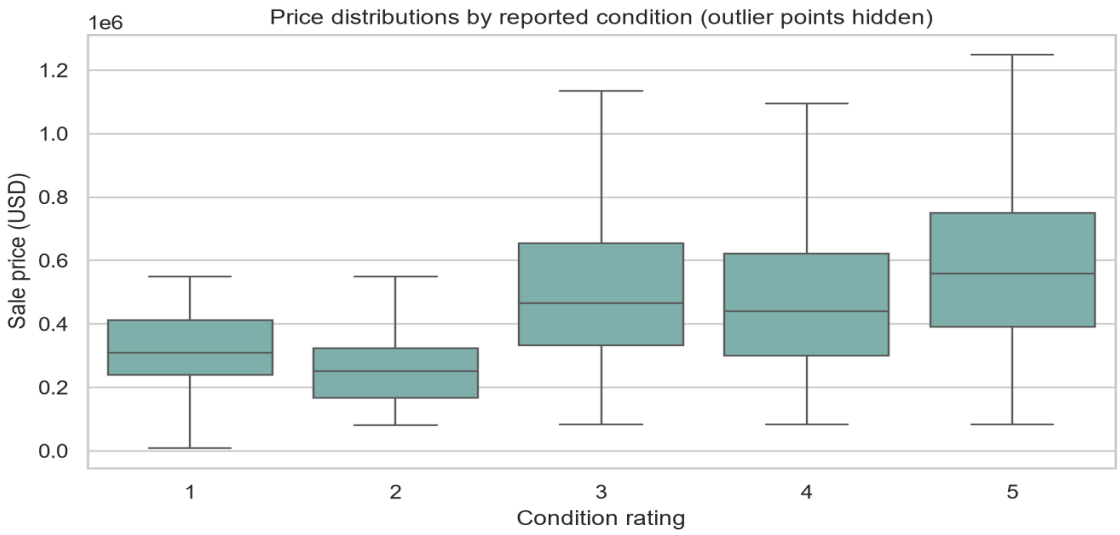
Numeric feature correlations



Several size-related measures move together, so correlations among features can make their separate relationships with price hard to interpret. Renovation year is coded as zero when no renovation year is recorded; it is not a normal continuous year for those rows.

4. Findings, limitations, and conclusion

Condition and sale price



Boxes show the middle half of prices within each condition rating; outlier points are hidden for readability. The groups overlap, so condition alone does not determine sale price.

Main findings

1. Of 4,600 raw records, 49 have a zero or negative price (1.1%) and were excluded from price statistics.
2. The positive-price median is $465,000, below the mean of $557,906, consistent with a right-skewed distribution.
3. Living area has a positive correlation with price (r = 0.45); this does not control for location or other home features.
4. Among cities with at least 20 positive-price sales, medians range from $238,500 (SeaTac) to $950,467 (Mercer Island). These are raw comparisons.
5. The dataset covers 02 May 2014 to 10 Jul 2014, too short a period for conclusions about long-run appreciation or current prices.

Limitations

The supplied CSV has no accompanying source citation, license, data dictionary, or collection-method notes. The reason for zero prices is unknown. Extreme observations are retained in the cleaned file rather than removed without evidence. City groups vary in size and mix. Reported home features may be incomplete, and sales are not adjusted for inflation or neighborhood-level differences. Findings are descriptive, not causal or predictive.

Conclusion

This compact dataset supports a clear EDA exercise: inspect quality, handle non-positive target values transparently, compare robust summaries, and interpret charts carefully. The strongest practical takeaway is that size and location show visible price patterns, while skew, data-quality questions, and narrow dates limit broader claims. The analysis should not be used to price an individual property.

Reproducibility and files

Run House_Price_EDA.ipynb from the project root. The notebook writes outputs/cleaned_house_prices.csv. The original supplied CSV is included in data/ for a self-contained project. Source: House_Price_data.csv, supplied by the project user; source metadata was not included with the file.